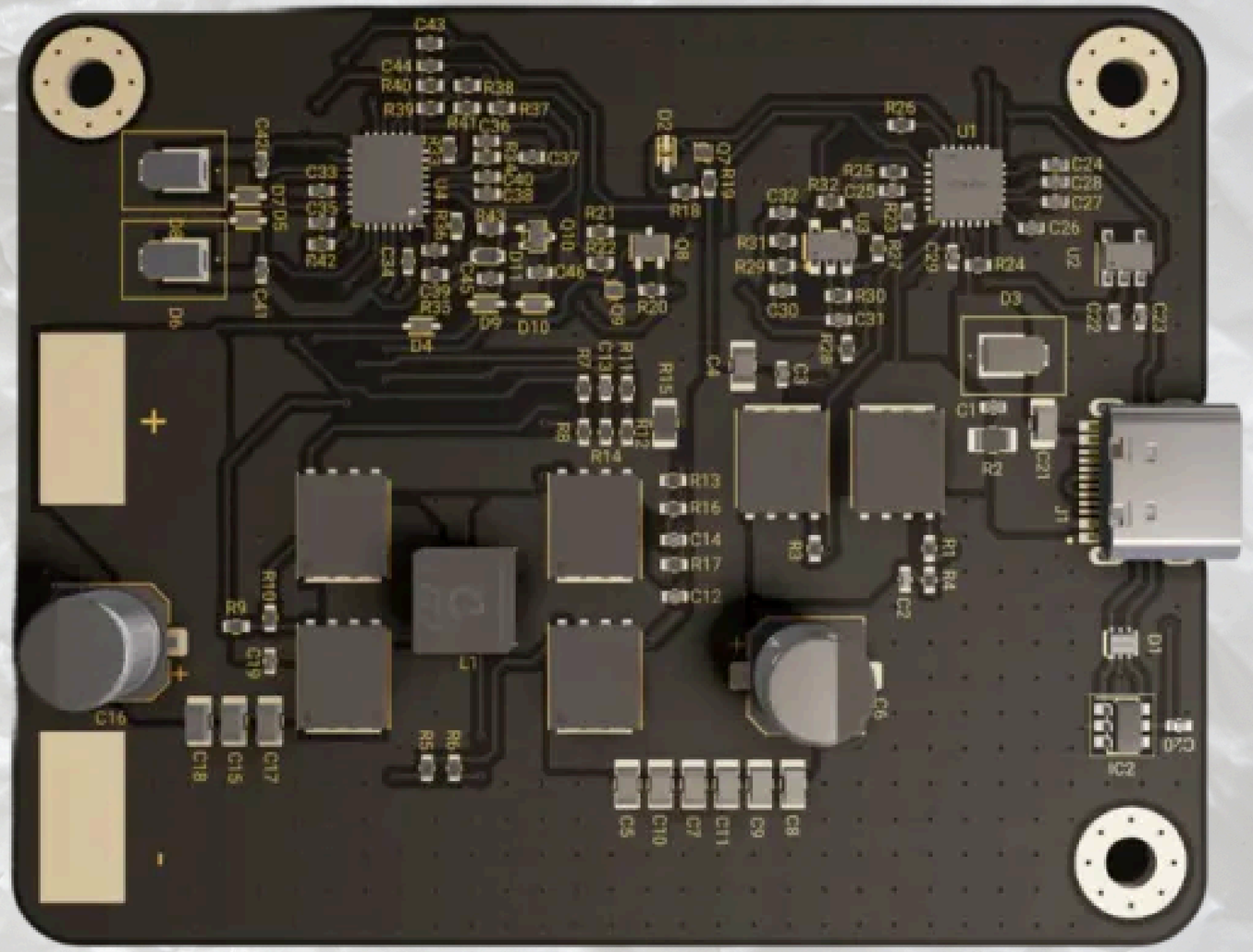
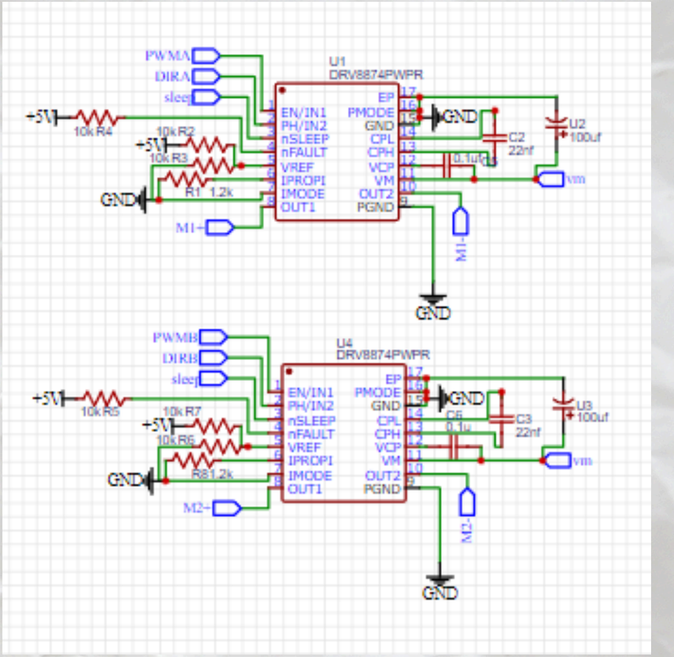
# Embedded Systems & PCB Design Intern AT Dyoraa Robotics



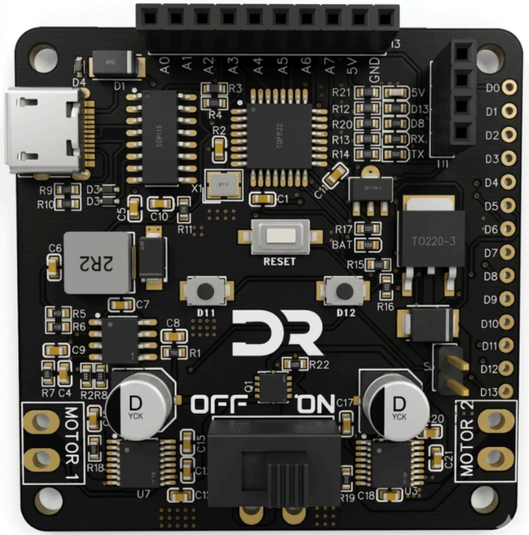
# About the company

- Focuses on PCB design, electronic circuitry, and embedded systems
- Develops custom electronic hardware for robotics and automation applications
- Works on microcontroller-based embedded systems and hardware integration
- Designs and develops sensor, motor-control, and power-management circuits
- Builds robotics prototypes and electronic systems from concept to working hardware
- Combines hardware design, embedded programming, and IoT technologies to develop practical solutions
- Emphasizes rapid prototyping, testing, debugging, and optimization of electronic systems



# What was my role?

- Designed PCBs and circuit schematics for robotics applications
- Programmed microcontrollers and embedded systems
- Integrated sensors, motors, motor drivers, and multiplexers
- Developed control and firmware logic for robotic systems
- Performed PCB assembly, soldering, testing, and debugging
- Designed power-management and motor-control circuits
- Worked with ROS2 and Gazebo for robotic simulation
- Contributed to robotic prototype development and optimization



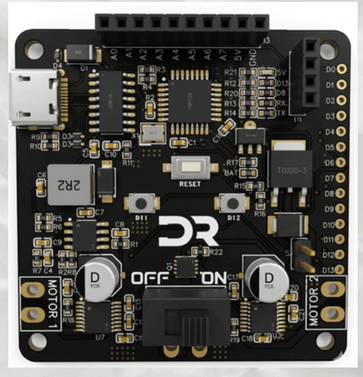
# What did i learn ?

- Learned PCB schematic design and layout
- Gained practical knowledge of embedded systems and microcontrollers
- Learned sensor and motor interfacing
- Developed skills in embedded firmware programming
- Learned PCB fabrication, SMD soldering, and assembly
- Gained experience in hardware testing and debugging
- Learned about power-management and motor-control circuits
- Worked with IoT-based hardware and communication systems
- Learned the basics of ROS2 and Gazebo for robotics
- Improved my understanding of hardware prototyping and system integration

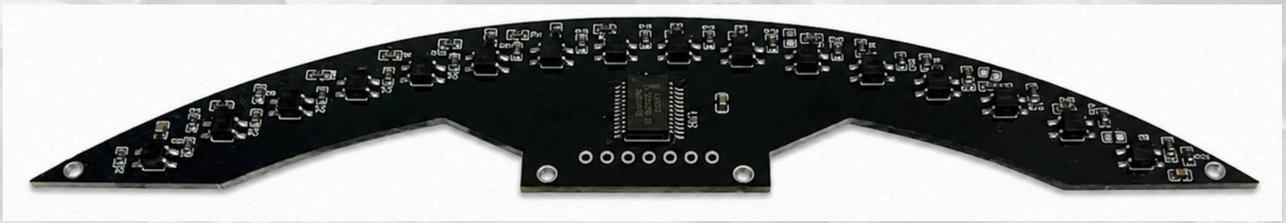
# My prototypes contributions




Full robot PCB
High Current capable




Robot carrier board

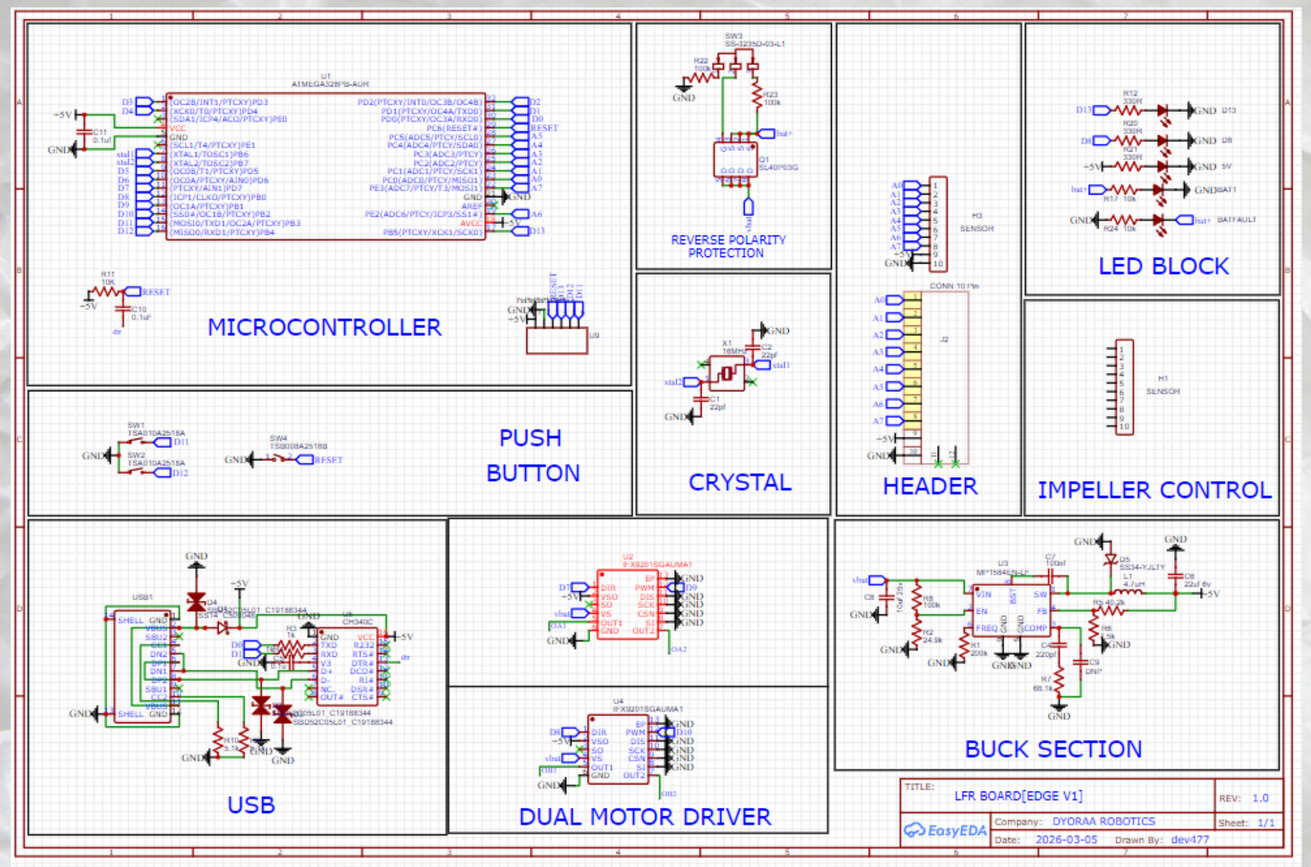


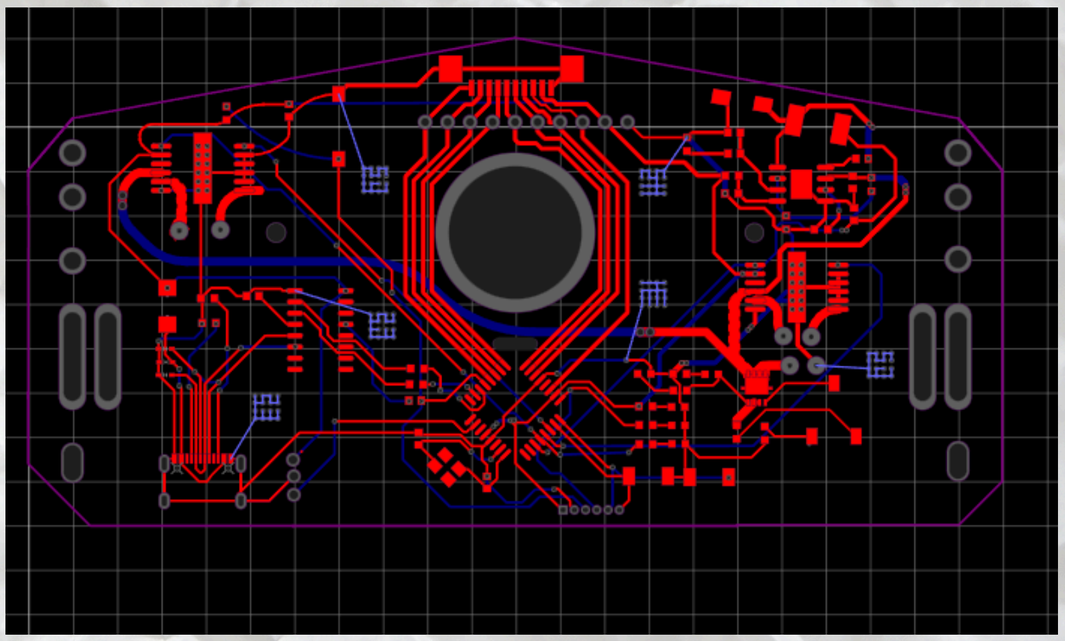
Robot sensor array

# Softwares i learned

# EasyEDA PRO

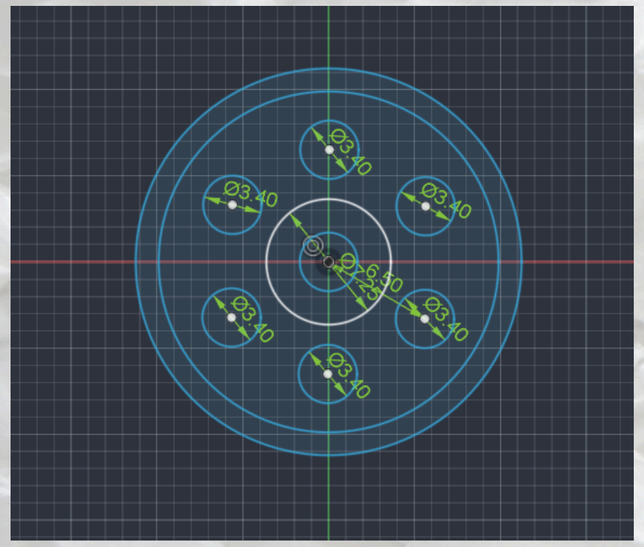
- EasyEDA is a web-based electronics design tool used for creating circuit schematics and PCB layouts. It provides component libraries, PCB routing, design-rule checking, and Gerber file generation, making it useful for designing, testing, and preparing PCBs for manufacturing.

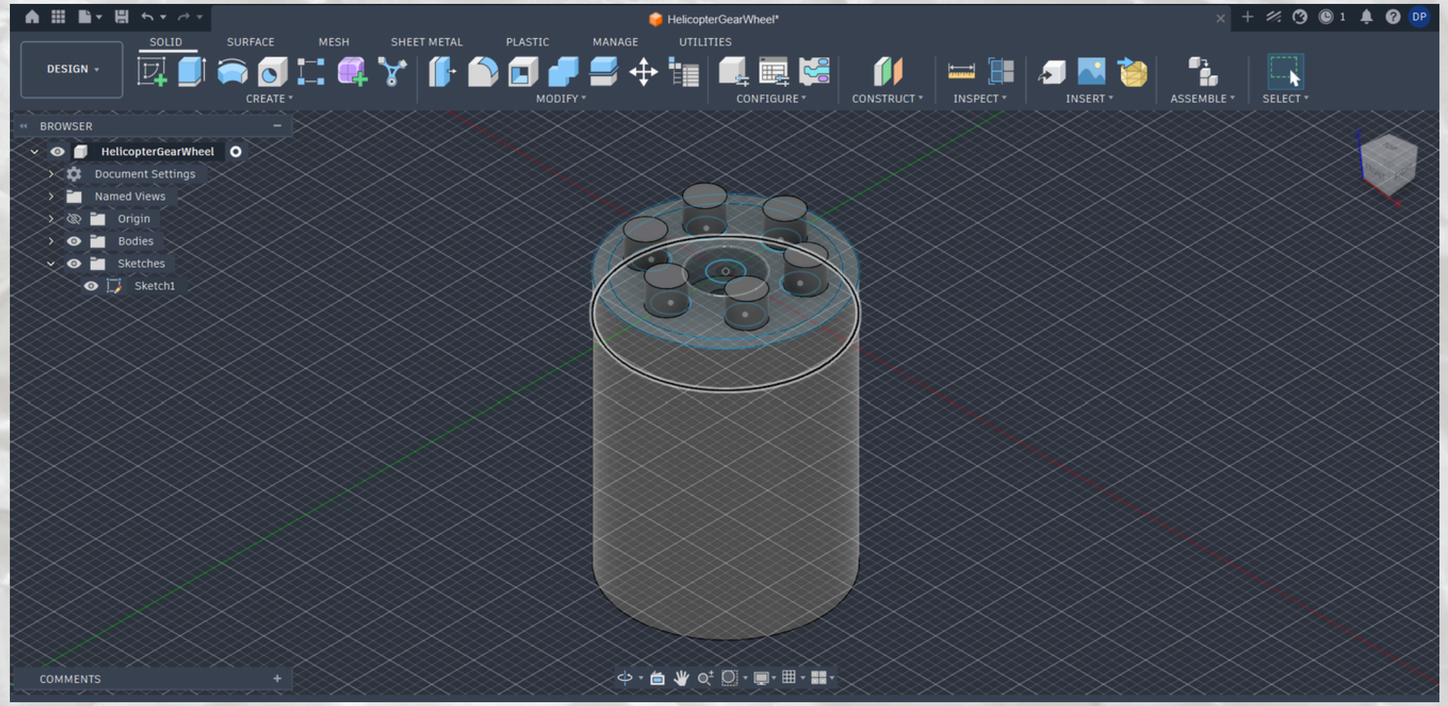


# AutoCad Fusion 360 Pro

- Autodesk Fusion is a 3D CAD and product design software used to create, modify, and assemble mechanical components. It provides tools for 3D modeling, assemblies, technical drawings, and design analysis, making it useful for developing mechanical parts and prototypes for robotics and engineering applications.



# Arduino Ide 2.3.6

- Arduino IDE is a development environment used to write, compile, and upload embedded programs to Arduino-compatible microcontrollers. It provides a simple interface for programming boards, testing hardware, and debugging code, making it useful for developing sensor-based, IoT, and robotic applications.

